Predicting NFL Play Types Using Pre-Snap Data: A Deep Learning Approach

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1 Introduction

Football is a game of inches and inches make the champion. This saying was cultivated by Vince Lombardi [1], who is regarded by many as the greatest football coach of all time. He had such a great influence on the game, that today the winner of the Super Bowl is lifting the Vince Lombardi Trophy.

In a game where every inch counts, being able to quickly process the available information dictates a team's success. This is especially important for a football team's defense. As they have to react to the play calls of the opposing offense, the defense's ability to correctly predict the type of play the offense is running can be a game-deciding advantage. On the highest level we distinguish two types of plays: A pass, where the Quarterback (QB) is throwing the football to one of his eligible receivers or a rush, where the ball carrier advances on the ground in order to gain yards towards the end zone. Each concept commands different defensive alignments, personnel packages and tactics. Within seconds defensive coaches must take the decision to commit to stopping one type of play while potentially exposing vulnerabilities to the other. Here, their ability to predict offensive intentions before the ball is snapped represents the difference between defensive success and failure.

While offenses are doing their best to disguise their play calls, pre-snap information such as formations and personnel packages can still give away what type of play is going to be run. These data, combined with contextual game state information - including the score, time remaining, down and field positioning - serve as cues to reveal offensive tendencies. This report examines how these cues can be used to predict an offense's upcoming play call.

2 Data Description

The data source used in this report is from the NFL Big Data Bowl 2025 competition [2]. It comprises detailed play-by-play data of 136 NFL games from Weeks 1-9 of the 2022 season. The data is split into multiple files containing information about games, individual plays from these games, players and individual player data for each respective play.

2.1 Play-level Information

The main data input for the model - plays.csv - contains play-level information from each game. While for each play 50 features are recorded, many of them are irrelevant for the prediction as they represent information that is not available to coaches before a play occurs - consequently these features were not used. Relevant input features can be split into two categories.

2.1.1 On-field pre-snap information

This includes the formation used by the offense (*offenseFormation*) as well as the alignment of the receivers (*receiverAlignment*). When looking at the visualizations of offense formations in Appendix A, it becomes apparent how this feature can help predict the play type. While for 'Empty' formations defenses can rather expect a pass play given that no Runningback (RB) is on the field (see 3e), in 'Wildcat' looks, where the ball is snapped directly to the RB (see 3f), a run play seems more imminent. This tendency is clearly illustrated in Figure 1 (for all visualizations

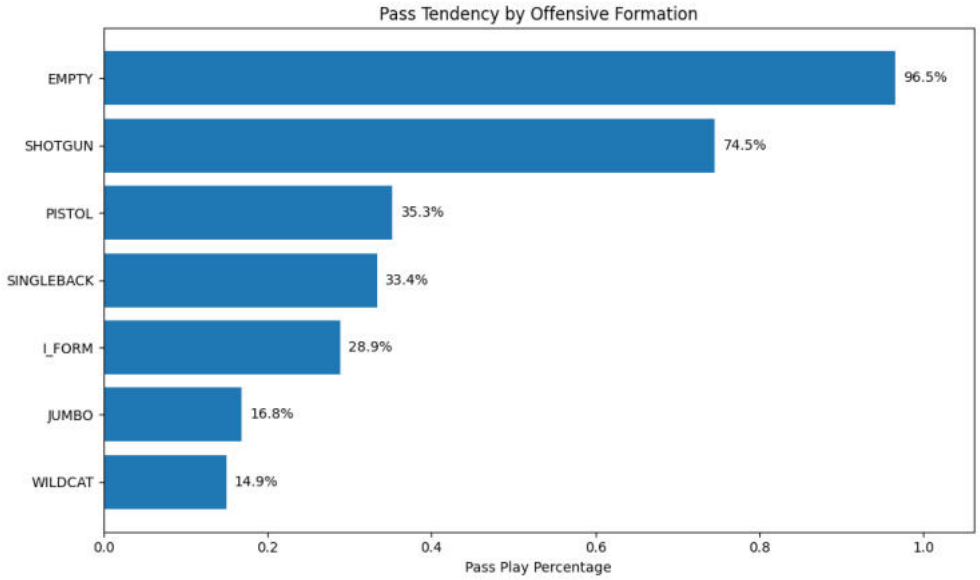

Figure 1: Relative Proportion of Pass plays for the different offensive Formations taken from play-by-play data

matplotlib was used [3]. Receiver Alignment gives insight into how many receivers are lined up on each side of the offensive line. The variable is encoded as $A \times B$, where A and B denote the number of receivers lined up on the right and left of the offensive line.
Along the existing data, additional features were derived to give a more complete look of the offense pre-snap. The *personnelPackage* provides a standardized representation of the offensive skill position composition using a common notation. This feature encodes the number of running backs and tight ends on the field in RB-TE format. Common packages include 11 personnel (1 RB, 1 TE), 12 personnel (1 RB, 2 TE), and 21 personnel (2 RB, 1 TE). This can serve as a strong predictor of offensive intent, as "heavier" packages with multiple Runningbacks and / or Tight Ends typically hint towards rushing plays, while lighter packages favor passing concepts.

Next to the skill position composition, the number of Offensive Linemen represents a critical component that directly influences play-calling tendencies. While the clear standard is 5 linemen (97.2%, see Figure 4), situational formations can deviate from this baseline. Goal-line and short-yardage situations often employ six or more linemen, creating additional blocking strength that strongly indicates rushing attempts. Together with formation and receiver alignment data, these derived features construct a comprehensive pre-snap profile of the offense.

2.1.2 Contextual Game State Information

Beyond the physical arrangement of players, the broader state of the game heavily influences the type of plays offensive coaches will call. Four variables of the main data input were relevant in this case. *Quarter* and *gameClock* represent the current quarter as well as the time remaining in that quarter. The temporal state of the game can influence play calling especially in combination with the score differential(see Table 2).

Down, *yardsToGo* and *absoluteYardLineNumber* further describe the field position of the offense. In football, offenses get 4 downs (plays) to advance 10 yards in order to receive a new set of downs. *Down* represents the current down with *yardsToGo* giving the distance needed for a first down - both can have a substantial influence on the play call. In short yardage situations ($\geq$ 3 yards to go) coaches will typically favor rush plays, whereas longer yardage situations ($\leq$ 8 yards to go) require higher-upside passing plays capable of generating substantial gains. The proportions in Table 1 show this kind of relationship. *AbsoluteYardLineNumber* concludes the information about field positioning, indicating the distance of the possession team to the endzone.

As an additional feature *scoreDifferential* was derived from the data, which is the score difference from the perspective of the possession team, i.e. if that team is currently behind, *scoreDifferential* is negative. As mentioned above, in later stages of a game, depending on the scoreDifferential, playcalls will vastly differ between teams. Trailing teams are incentivised to call pass plays in order to quickly drive down the field and score points, whereas leading

Table 1: Pass Play Proportion by Down and Yardage Situation

Down	**Short ($\geq$ 3 yds)**	**Medium (4-7 yds)**	**Long ($\leq$ 8 yds)**
1	0.327	0.333	0.513
2	0.399	0.563	0.733
3	0.515	0.911	0.932
4	0.571	0.982	0.976

teams will more often call rushing plays to drain the clock and avoid turnovers. Table 2 confirms this.

Table 2: Pass Play Proportion by Quarter and Game State

Quarter	**Leading Team**	**Trailing Team**
1	0.557	0.600
2	0.651	0.644
3	0.552	0.600
4	0.387	0.768

2.2 Additional Information

Other files containing game and player as well as player-level play data were not used as features themselves but merely for feature engineering.

2.3 Target Variable

The target variable for the prediction is *isDropback*. It indicates, whether the QB dropped back to pass the ball (1) or not (0). This is important for our case, as it focuses on whether it was the plan of the offense to pass the ball, not wether a pass actually occurred. Excluded from the dataset were QB Kneels and Spikes, identified by *qbKneel* and *qbSpike* respectively. Kneels were classified as non-dropbacks (rush plays), Spikes as dropbacks (pass plays), but as for both of these plays the offense's intention is neither to rush nor pass the ball, they were removed. After removal, the dataset contains 15 936 plays, with 60,95% classified as pass plays. This percentage serves as a baseline comparison for our model's performance, as naively predicting a pass on every play would give exactly this accuracy.

3 Method Description

3.1 Method Selection

As our target variable is binary we are looking at a binary classification task. While this opens up multiple analysis approaches, given the large scale of the data, a neural network was chosen here. Neural networks are able to reach high classification accuracy and have shown in related work to be the most accurate in predicting play types [4] [5]. Targeting a high prediction accuracy via a neural network furthermore makes this work more relevant for application purposes - according to the Head of Data Science of the New England Patriots, the focus of play prediction should be high accuracy as coaches require data-driven insights to be practically guaranteed to work to incorporate such models into their play-calling [4]. They are also best equipped to working with the data at hand. From section 2 it becomes clear that many of the features are non-linearly correlated. While e.g. a Logistic Regression would require manual specification of these interaction terms, neural networks automatically learn such complex relationships. The network was built using *pytorch* [6].

3.2 Feature Processing

Before feeding them to the network, the features still needed to be processed. As a truly categorical variable, *offenseFormation* was transformed using the OneHotEncoder class from *sklearn* [7]. While the variables *receiverAlignment* and *personnelPackage* can be considered categorical, they have ordinal properties that would have been lost through one-hot encoding. Therefore, they were split into their numerical components. The feature *gameClock*

was transformed from its format (MM:SS) into a true numerical form, showing the game clock in seconds. To avoid larger numerical magnitudes to dominate the model and speed up the training process, all numerical features were scaled. Here, the standard scaler was used - which standardized features by removing the mean and scaling to unit variance - as it led to slightly better accuracy. The dataset was then split into 70% training, 10% validation, and 20% test sets, with stratification to ensure equal distribution of pass/rush plays across splits. For the feature processing as well as the feature engineering the python libraries *pandas* [8] [9] and *numpy* [10] were used.

3.3 Model Architecture

For the network architecture design, a standard multi-layer perceptron (MLP) was chosen. While other architectures were considered, they were ultimately ruled out given the nature of the available data. Since individual plays are independent observations without temporal dependencies - at least across the whole dataset - recurrent neural networks were disregarded and because the feature space did not consider exact pre-snap player positioning on the field (x, y coordinates), more complex architectures such as graph neural networks or attention mechanisms, which would be valuable for analyzing spatial relationships, were not applicable. Binary Cross Entropy (BCE) loss was selected as the loss function, which is the standard choice for binary classification problems and the Adam optimizer was used for training with a learning rate of 0.001 and weight decay of 0.0001. The MLP model is a 4-layer fully connected neural network with ReLU activation functions in hidden layers, dropout regularization and sigmoid activation for the final binary output.

3.4 Challenges

The biggest challenge during model training was managing overfitting. At the beginning the model suffered heavily from this issue, with training and validation scores diverging as early as after 7 epochs, limiting model accuracy to around 70%. To counteract this, several measures were introduced. Dropout layers were added to the model and the weight decay of 0.0001 was applied to the Adam optimizer to regularize the weights of the network. Additionally, early stopping with a patience of 5 epochs was implemented to halt training when validation loss stopped improving. These measures collectively increased model performance by approximately 5%, bringing the final accuracy to around 75%.

4 Evaluation

The main focus of model evaluation was accuracy, as that seems to be most relevant for real-world application (see 3.1). The model achieved 75% accuracy, consistent with the performance reported in related work [4] [5], demonstrating the model's competitiveness against established benchmarks. The learning curves in Figure 2 reveal that validation loss began diverging from training loss quite early, validating the importance of the implemented regularization techniques.

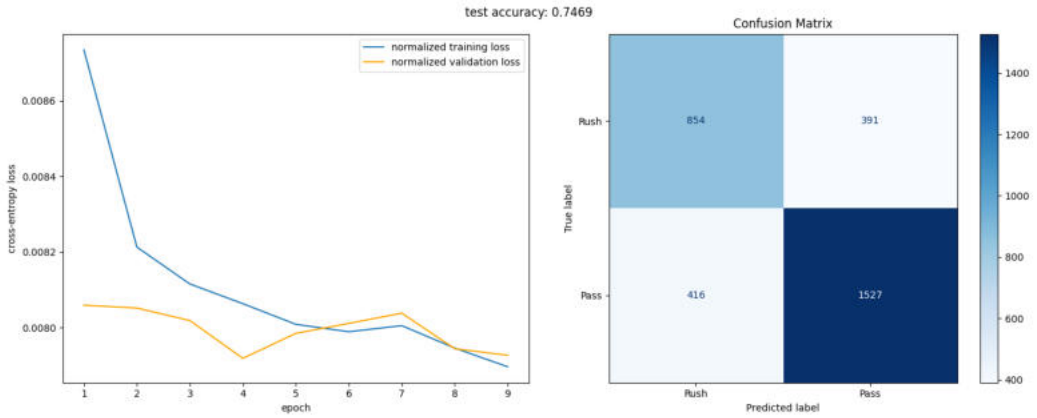

Figure 2: Learning curve and confusion matrix of the Model

Analyzing the confusion matrix in Figure 2, we see that the model has demonstrated better performance predicting pass plays (80% sensitivity) compared to rush plays (67% specificity). This can have multiple reasons. Our dataset

reflected the NFL's pass-heavy nature with 61% pass plays and 39% rush plays. As training and validation splits were done to follow this distribution, during training the model was provided with more pass plays, potentially leading to better predictive performance. It could also suggest that our features offer clearer indicators for passing plays than rushing plays, making them inherently more challenging to predict from the data at hand.

5 Future Work

Several promising directions exist for extending this analysis. The data source [2] used in this report also provides detailed spatial information including player positions (x, y coordinates), speed, and motion direction for each player on a per-frame basis. Incorporating this data with models capable of learning spatial interactions between players - such as transformer architectures utilizing attention mechanisms - would likely yield more precise predictions and deeper insights into play-calling patterns. Ranasaria and Vabishchevich provide an interesting framework for applying transformer architecture to sports tracking data [11].

Additionally, incorporating player performance metrics such as Pro Football Focus (PFF) grades for key skill position players (quarterbacks, running backs, wide receivers) could further enhance prediction accuracy, as related work has demonstrated the influence of individual player capabilities on play-calling decisions [4]. Such data was unfortunately not provided by the data source.

6 Conclusion

This work aimed at predicting NFL play types from pre-snap information using deep learning techniques. A four-layer neural network achieved 75% accuracy on pass/rush classification, in line with comparable work. Evaluating model performance revealed a significant disparity between pass prediction accuracy (80% sensitivity) and rush prediction accuracy (67% specificity), indicating that rush plays remain more challenging to predict from pre-snap information alone. Future work should incorporate analyzing spatial player relationships through advanced architectures and use improved feature sets to investigate whether this performance disparity can be addressed.

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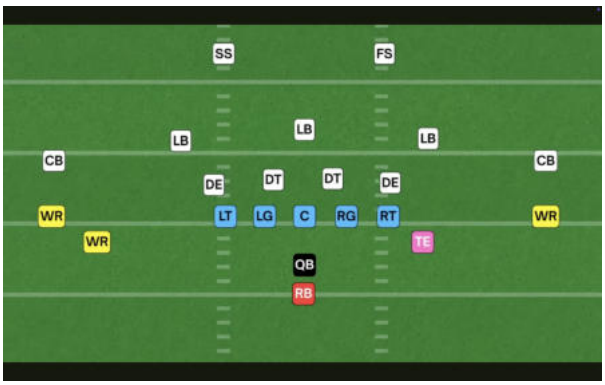
A Offense Formations Visualization

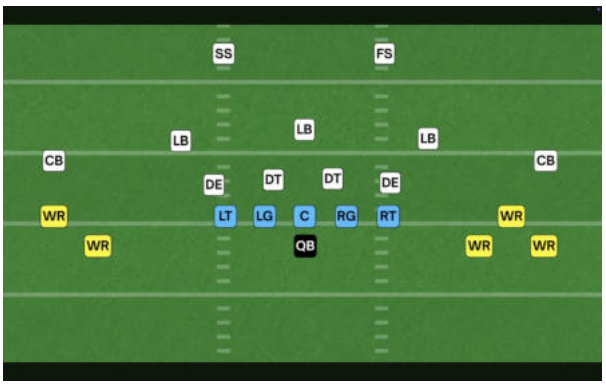
(a) I-Formation

(b) Singleback

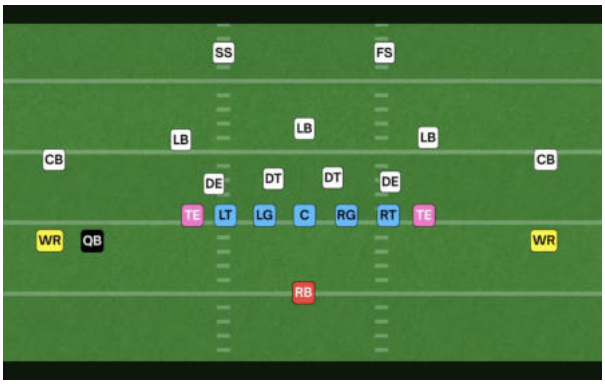
(c) Shotgun



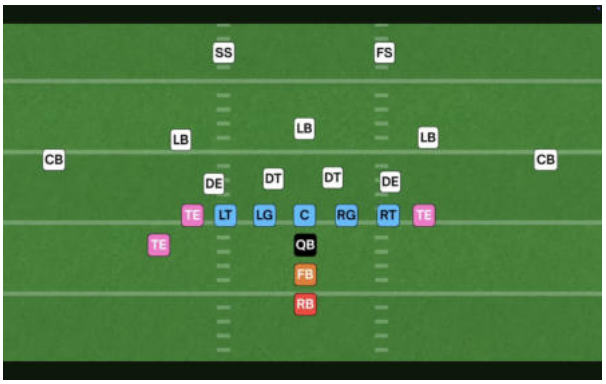
(d) Pistol



(e) Empty



(f) Wildcat



(g) Jumbo

Figure 3: NFL Offensive Formations. Visualizations taken from [12] for the different values that the feature *offenseFormation* can take.

B Distribution of the number of Linemen per play

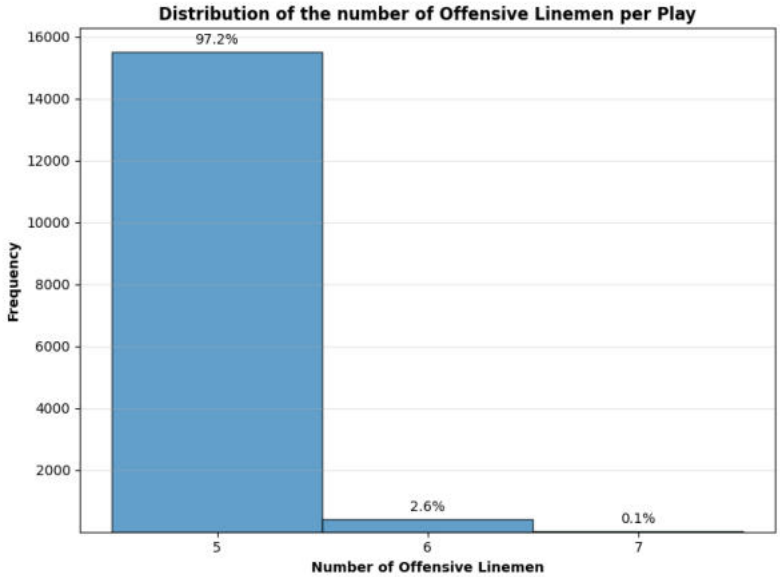


Figure 4: Distribution of the number of Offensive Linemen used per Play